Water Filtration Engineering Challenge

The University of Oklahoma

Name: _____

Date: _____

Rapid Sand Filtration Test

Materials:

- Conical tubes (50ml)
- Sand (40 x 50)
- Small rock (60 x 80)
- Medium rock (80 x 120)
- Large rock (1/4 x 1/8)
- Funnel with PVC tubing attachment
- 1 liter of dirty sample water
- 1 liter plastic container (x2)
- Weigh boats
- Mass balance

Practice Laboratory:

1. At your filter testing station, add 20 ml of CLEAN water inside your conical tube.
2. Begin adding the media to your column in the corresponding order. **NOTE: The media will need to be weighed using the mass balance. See mass balance instructions.**

Order	Amount	Media	Notes:
1	5 grams	1/4 x 1/8	Rinse the conical tubes using the "DI" water containers to remove any excess media each time.
2	2.5 grams	80 x 120	
3	5 grams	40 x 50	
4	7.5 grams	60 x 80	
5	5 grams	80 x 120	

3. After all your media is added, add an additional 50 ml of CLEAN water to your conical tubes and place it under your filter. Allow the water to filter through the bottom.
4. Once the water level is right above the media, plug the bottom of the conical tube with paper towels. **DO NOT LET MEDIA DRY!**
5. Prepare a mixture of dirty water in a beaker using dirt and water.
6. Take out a sample and measure the **initial** clarity/turbidity.
    a. **Note: Clarity can be measured by an instrument called a turbidimeter. See turbidimeter instructions. Record the initial value from the turbidimeter in the table below.**
7. Once you've measured the turbidity, pour the entire contents into your conical tube and remove paper towel. Allow the dirty sample water to filter through.
8. Measure the **final** turbidity once your conical tube has finished filtering.
    a. **See turbidimeter instructions. Record value in table below.**

	Initial Turbidity	Final Turbidity
Column Test		

Part 1: Experimenting

In part 1, you will be allowed to free test using the 50 ml conical tubes and the different rock media available. The conical tubes will be used to prepare different filter media combinations which could be used as your final design.

Use this time to test out different variations of media to create your unique filter. In addition, consider advantages and disadvantages of each design as well as cost and time to build and run your filters. **See cost analysis sheet to see the cost of your capital and operation and maintenance of your filter.** Be sure to test your different filters for initial and final turbidity, time, and odor (smell test).

Your objective is to reduce the turbidity to 1 NTU. You must have the final turbidity 1 NTU or less for 1 liter of sample water with lowest cost and maximum volume of filter media. Using your lab notebook, create a table for your different variations of filter media. Here is an example of a table:

Trials	Initial Turbidity	Final Turbidity	Time	Odor	Description: Media Order
1					
2					

Part 2: Constructing & Testing

1. At your filtering station, make sure your valve to your filter is closed.
2. Using your plastic container, add 50 ml of CLEAN water inside your column or fill until it reaches the line indicated on your column.
3. Begin adding your media to your column in the order your team has chosen. **NOTE: The media will need to be weighed using the mass balance. See mass balance instructions.**
4. After all your media is added, add an additional 200 ml of CLEAN water to your filter using your plastic container and place it under your filter. Allow the water to filter through until it reaches the line indicated on your column.
5. Close the valve once you start to see the water level in your column and pour the contents of your plastic container into the "waste" container. **DO NOT LET MEDIA DRY!**
6. Retrieve your 1-liter bottle of DIRTY water and shake well. Measure the **initial** turbidity and write your result.
    a. **See turbidimeter instructions. Record value in table below.**
7. Using your 1-liter bottle of DIRTY water, pour the entire contents into your filter and start your timer. Place plastic container under your filter.
8. Open the valve and allow to filter through.
9. Test for turbidity when your plastic container reaches 200ml.
10. Test for turbidity when your plastic container reaches 400 ml.
11. Test for turbidity when your plastic container reaches 800ml.
12. Once your plastic container reaches 800ml of filtered water, pour it into your "waste" container.
13. With the remaining 200ml of filtered water in your plastic container, measure the **final** turbidity (at 1 L). Close the valve once you see the water level in the column.
    a. **See turbidimeter instructions. Record value in table below.**
14. In your lab notebook, write down your final time and turbidity results like the table below.

	Initial Turbidity (NTU) @ 200 ml	Turbidity (NTU) @ 400 ml	Turbidity (NTU) @ 600 ml	Turbidity (NTU) @ 800 ml	Final Turbidity (NTU) @ 1 L	Time (min)
Column Test						

Part 3: Testing Analysis

1. Prepare a table showing your cost analysis and filter run time for your filter. *See cost analysis page 5.*
2. Create a bar graph to show turbidity performances for **each** of your trials from part 1 of experimentation.
    a. **NOTE: Be sure to have all trials on the same graph**
3. Create another bar graph showing the time of **each** trial filter prepared in part 1 of experimentation.
    a. **NOTE: Be sure to have all trials on the same graph**
4. Prepare a bar graph showing your actual design turbidity results from your filter in part 2 of the laboratory.
    a. **NOTE: Be sure to have initial turbidity and final turbidity**

Part 4: Presentation

Prepare a 3 – minute presentation that includes the following:

- Selected media (mass and grain size)
- Media order
- Total cost
- Prepared tables and graphs
- Final effluent turbidity

In your presentation, be sure to discuss your thought process. What were some advantages and disadvantages you noticed? How could you improve your filter?

Be creative! Come up with a team name! Add pictures, videos, graphics, clip art, animations, and transitions. Also, be sure to label all tables and graphs.

Cost Analysis

Subject	*Conditions*	*Equations*	*Variables*
Filter Run Time Cost (O&M)	$0 < \mathbf{A} < 30$: $\mathbf{A} \geq 30$:	$\mathbf{Z_1} = \$10.00$ $\mathbf{Z_1} = \mathbf{A}\text{-}20$	**A = time** **Z_1 = cost**
Capital Cost for Each Media	Sand (40 x 50) Small rock (60 x 80) Medium Rock (80 x 120) Large Rock (80 x 120)	$\mathbf{Z_2} = \$2.00$/gram $\mathbf{Z_2} = \$1.75$/gram $\mathbf{Z_2} = \$1.25$/gram $\mathbf{Z_2} = \$1.00$/gram	**ΣZ_2 = cost**
Turbidity Penalty Cost	$\mathbf{T} < 1$: $\mathbf{T} > 50$: $1 < \mathbf{T} < 50$:	$\mathbf{Z_3} = \$0.00$ $\mathbf{Z_3} = \$100.00$ $\mathbf{Z_3} = 1.75\mathbf{T}$	**T = Turbidity** **Z_3 = Cost**

	Final Costs
Filter Run Time Cost **(Z_1)**	$
Capital Cost for Each Media **(Sum of Z_2 = ΣZ_2)**	$
Turbidity Penalty Cost **(Z_3)**	$
SUM ($Z_1 + \Sigma Z_2 + Z_3$)	$